United States Patent	12417074
Kind Code	B1
Date of Patent	September 16, 2025
Inventor(s)	Carasso; R. David et al.

Updating event records based on user edited extraction rule

Abstract

Embodiments are directed towards real time display of event records with an indication of previously provided extraction rules. A plurality of extraction rules may be provided to the system, such as automatically generated and/or user created extraction rules. These extraction rules may include regular expressions. A plurality of event records may be displayed to the user, such that text in a field defined by an extraction rule is emphasized in the display of the event record. The same emphasis may be provided for text in overlapping fields, or the emphasis may be somewhat different for different fields. The user interface may enable a user to select a portion of text of an event record, such as by rolling-over or clicking on an emphasized part of the event record. By selecting the portion of the event record, the interface may display each extraction rule associated with the selected portion.

Inventors: **Carasso; R. David (San Rafael, CA), Delfino; Micah James (San Francisco, CA), Hwang; Johnvey (San Francisco, CA)**

Applicant: **Splunk Inc.** (San Francisco, CA)

Family ID: **1000007566039**

Appl. No.: **18/457797**

Filed: **August 29, 2023**

Related U.S. Application Data

continuation parent-doc US 17384467 20210723 US 11782678 child-doc US 18457797
continuation parent-doc US 17028755 20200922 US 11119728 20210914 child-doc US 17384467
continuation parent-doc US 16003998 20180608 US 10802797 20201013 child-doc US 17028755
continuation parent-doc US 14266839 20140501 US 10019226 20180710 child-doc US 16003998
continuation parent-doc US 13748391 20130123 US 8751963 20140610 child-doc US 14266839

Publication Classification

Int. Cl.: **G06F3/048** (20130101); **G06F7/24** (20060101); **G06F16/2458** (20190101)

U.S. Cl.:

CPC **G06F7/24** (20130101); **G06F16/2477** (20190101);

Field of Classification Search

CPC: G06F (7/24); G06F (16/247)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5550971	12/1995	Brunner et al.	N/A	N/A
5761389	12/1997	Maeda et al.	N/A	N/A
5764975	12/1997	Taniguchi et al.	N/A	N/A
5913032	12/1998	Schwartz et al.	N/A	N/A
6049777	12/1999	Sheena et al.	N/A	N/A
6112186	12/1999	Bergh et al.	N/A	N/A
6118936	12/1999	Lauer et al.	N/A	N/A
6173418	12/2000	Fujino et al.	N/A	N/A
6208720	12/2000	Curtis et al.	N/A	N/A
6311194	12/2000	Sheth et al.	N/A	N/A
6347374	12/2001	Drake et al.	N/A	N/A
6374251	12/2001	Fayyad et al.	N/A	N/A
6549208	12/2002	Maloney et al.	N/A	N/A
6609128	12/2002	Underwood	N/A	N/A
6718535	12/2003	Underwood	N/A	N/A
6839669	12/2004	Gould et al.	N/A	N/A
6954756	12/2004	Arning et al.	N/A	N/A
7035925	12/2005	Nareddy et al.	N/A	N/A
7085682	12/2005	Heller et al.	N/A	N/A
7100195	12/2005	Underwood	N/A	N/A
7136880	12/2005	Wilkins et al.	N/A	N/A
7219239	12/2006	Njemanze et al.	N/A	N/A
7376969	12/2007	Njemanze et al.	N/A	N/A
7389306	12/2007	Schuetze et al.	N/A	N/A
7503012	12/2008	Chen et al.	N/A	N/A
7562069	12/2008	Chowdhury et al.	N/A	N/A
7644414	12/2009	Smith et al.	N/A	N/A
7650512	12/2009	Karimisetty et al.	N/A	N/A
7650638	12/2009	Njemanze et al.	N/A	N/A
7702639	12/2009	Stanley et al.	N/A	N/A
7769899	12/2009	Grabarnik et al.	N/A	N/A
7779021	12/2009	Smith et al.	N/A	N/A
7805482	12/2009	Schiefer	N/A	N/A

7809131	12/2009	Njemanze et al.	N/A	N/A
7899783	12/2010	Xu et al.	N/A	N/A
7950062	12/2010	Ren et al.	N/A	N/A
7958164	12/2010	Ivanov et al.	N/A	N/A
8022987	12/2010	Ko et al.	N/A	N/A
8112398	12/2011	Hernandez	N/A	N/A
8121973	12/2011	Anderson et al.	N/A	N/A
8200506	12/2011	Kil	N/A	N/A
8412696	12/2012	Zhang et al.	N/A	N/A
8442950	12/2012	D'Souza et al.	N/A	N/A
8442982	12/2012	Jacobson et al.	N/A	N/A
8458612	12/2012	Chatterjee et al.	N/A	N/A
8516008	12/2012	Marquardt et al.	N/A	N/A
8553993	12/2012	Flesselles et al.	N/A	N/A
8578500	12/2012	Long	N/A	N/A
8589403	12/2012	Marquardt et al.	N/A	N/A
8682925	12/2013	Marquardt et al.	N/A	N/A
8700658	12/2013	Rambhia et al.	N/A	N/A
8713000	12/2013	Elman et al.	N/A	N/A
8751855	12/2013	Yairi et al.	N/A	N/A
8752178	12/2013	Coates et al.	N/A	N/A
8788526	12/2013	Neels et al.	N/A	N/A
8806361	12/2013	Noel et al.	N/A	N/A
8826434	12/2013	Merza	N/A	N/A
8965889	12/2014	Chan et al.	N/A	N/A
9032281	12/2014	Zongker	N/A	N/A
9037667	12/2014	Rivkin	N/A	N/A
9077715	12/2014	Satish et al.	N/A	N/A
9124612	12/2014	Vasan et al.	N/A	N/A
9130971	12/2014	Vasan et al.	N/A	N/A
9185007	12/2014	Fletcher et al.	N/A	N/A
9189064	12/2014	Chaudhri et al.	N/A	N/A
9215240	12/2014	Merza et al.	N/A	N/A
9262357	12/2015	Accapadi et al.	N/A	N/A
9269095	12/2015	Chan et al.	N/A	N/A
9270518	12/2015	Muro et al.	N/A	N/A
9292361	12/2015	Chitilian et al.	N/A	N/A
9298854	12/2015	Ikawa et al.	N/A	N/A
9363149	12/2015	Chauhan et al.	N/A	N/A
9516052	12/2015	Chauhan et al.	N/A	N/A
9582557	12/2016	Carasso et al.	N/A	N/A
9588667	12/2016	Singhal et al.	N/A	N/A
9740788	12/2016	Blank, Jr. et al.	N/A	N/A
9747010	12/2016	Prabhat et al.	N/A	N/A
9778834	12/2016	Evans et al.	N/A	N/A
9798952	12/2016	Kawazu	N/A	N/A
9864797	12/2017	Fletcher et al.	N/A	N/A
9875319	12/2017	Wolfram et al.	N/A	N/A
9904456	12/2017	Xie et al.	N/A	N/A
9922084	12/2017	Robichaud	N/A	N/A

9922099	12/2017	Lamas et al.	N/A	N/A
9923767	12/2017	Dickey	N/A	N/A
9967351	12/2017	Maheshwari et al.	N/A	N/A
9977803	12/2017	Robichaud et al.	N/A	N/A
9996446	12/2017	Lefor et al.	N/A	N/A
10009391	12/2017	Smith et al.	N/A	N/A
10019496	12/2017	Bingham et al.	N/A	N/A
10026045	12/2017	Portnoy et al.	N/A	N/A
10031905	12/2017	Fu et al.	N/A	N/A
10061832	12/2017	Brown et al.	N/A	N/A
10127255	12/2017	Campbell et al.	N/A	N/A
10127258	12/2017	Lamas et al.	N/A	N/A
10225136	12/2018	Bingham et al.	N/A	N/A
10229150	12/2018	Marquardt et al.	N/A	N/A
10248739	12/2018	Hamada	N/A	N/A
10318541	12/2018	Bingham et al.	N/A	N/A
10353957	12/2018	Bingham et al.	N/A	N/A
10387396	12/2018	Marquardt et al.	N/A	N/A
10409794	12/2018	Marquardt et al.	N/A	N/A
10417120	12/2018	Maag et al.	N/A	N/A
10474674	12/2018	Marquardt et al.	N/A	N/A
10565220	12/2019	Porath et al.	N/A	N/A
10585788	12/2019	Nallabothula et al.	N/A	N/A
10614132	12/2019	Bingham et al.	N/A	N/A
10621762	12/2019	Donalek et al.	N/A	N/A
10672161	12/2019	Maruyama et al.	N/A	N/A
10698896	12/2019	Halterman et al.	N/A	N/A
10771486	12/2019	Murphey et al.	N/A	N/A
10775976	12/2019	Abdul-Jawad et al.	N/A	N/A
10776441	12/2019	Echeverria et al.	N/A	N/A
10778712	12/2019	Chauhan et al.	N/A	N/A
10783195	12/2019	Musuluri	N/A	N/A
10795555	12/2019	Burke et al.	N/A	N/A
10832456	12/2019	Levy	N/A	N/A
10848510	12/2019	Chauhan et al.	N/A	N/A
10885450	12/2020	Kaitha	N/A	N/A
10896205	12/2020	Krishna et al.	N/A	N/A
10915338	12/2020	Lawrence	N/A	N/A
10915583	12/2020	Robichaud et al.	N/A	N/A
10922083	12/2020	Balasubramanian et al.	N/A	N/A
10922341	12/2020	De Boer	N/A	N/A
10922493	12/2020	Das et al.	N/A	N/A
10922660	12/2020	Rakshit et al.	N/A	N/A
10922892	12/2020	Bhushan et al.	N/A	N/A
10942774	12/2020	Aleti et al.	N/A	N/A
10985970	12/2020	Goyal et al.	N/A	N/A
10997180	12/2020	James et al.	N/A	N/A
10997190	12/2020	Porath et al.	N/A	N/A
10997192	12/2020	Boster et al.	N/A	N/A
10999164	12/2020	Sridhar et al.	N/A	N/A

11061918	12/2020	Miller et al.	N/A	N/A
11138191	12/2020	De Boer	N/A	N/A
11151125	12/2020	Dwivedi et al.	N/A	N/A
11194564	12/2020	Dwivedi et al.	N/A	N/A
11222066	12/2021	Batsakis et al.	N/A	N/A
11226964	12/2021	Cairney et al.	N/A	N/A
11231840	12/2021	Burke et al.	N/A	N/A
11232100	12/2021	Bhattacharjee et al.	N/A	N/A
11245581	12/2021	Hsiao et al.	N/A	N/A
11263229	12/2021	Basavaiah et al.	N/A	N/A
11269876	12/2021	Basavaiah et al.	N/A	N/A
11327827	12/2021	Satish et al.	N/A	N/A
11386133	12/2021	Neels et al.	N/A	N/A
11507562	12/2021	Leudtke et al.	N/A	N/A
11526504	12/2021	Moshgabadi et al.	N/A	N/A
11537942	12/2021	Vogler-ivashchanka et al.	N/A	N/A
11567993	12/2022	Batsakis et al.	N/A	N/A
11574242	12/2022	Burke et al.	N/A	N/A
11609913	12/2022	Anwar et al.	N/A	N/A
11620303	12/2022	Roy et al.	N/A	N/A
11663109	12/2022	Deaderick et al.	N/A	N/A
11782920	12/2022	Wang et al.	N/A	N/A
12038926	12/2023	Pathak	N/A	G06F 16/2228
12072939	12/2023	Batsakis	N/A	G06F 16/9032
12093272	12/2023	Batsakis	N/A	G06F 16/2471
12099492	12/2023	Bagga	N/A	H04L 67/146
12106419	12/2023	Bhushan	N/A	G06T 15/04
12197451	12/2024	Haggie	N/A	G06F 16/904
12242495	12/2024	Bolognese	N/A	G06F 16/2477
2001/0032205	12/2000	Kubaitis	N/A	N/A
2002/0049740	12/2001	Arning et al.	N/A	N/A
2002/0049838	12/2001	Sylor et al.	N/A	N/A
2002/0054101	12/2001	Beatty	N/A	N/A
2002/0133513	12/2001	Townsend et al.	N/A	N/A
2003/0061212	12/2002	Smith et al.	N/A	N/A
2003/0115333	12/2002	Cohen et al.	N/A	N/A
2003/0120475	12/2002	Nakamura	N/A	N/A
2003/0126056	12/2002	Hausman et al.	N/A	N/A
2003/0167192	12/2002	Santos et al.	N/A	N/A
2003/0187821	12/2002	Cotton et al.	N/A	N/A
2003/0220940	12/2002	Futoransky et al.	N/A	N/A
2004/0001049	12/2003	Bradley et al.	N/A	N/A
2004/0078359	12/2003	Bolognese et al.	N/A	N/A
2004/0133566	12/2003	Ishiguro et al.	N/A	N/A
2004/0220965	12/2003	Harville et al.	N/A	N/A
2004/0225641	12/2003	Dettinger et al.	N/A	N/A
2004/0243614	12/2003	Boone et al.	N/A	N/A
2004/0254919	12/2003	Giuseppini	N/A	N/A
2005/0015624	12/2004	Ginter et al.	N/A	N/A

2005/0022207	12/2004	Grabarnik et al.	N/A	N/A
2005/0108211	12/2004	Karimisetty et al.	N/A	N/A
2005/0108283	12/2004	Karimisetty et al.	N/A	N/A
2005/0114707	12/2004	DeStefano et al.	N/A	N/A
2005/0160086	12/2004	Haraguchi et al.	N/A	N/A
2005/0172162	12/2004	Takahashi et al.	N/A	N/A
2005/0203876	12/2004	Cragun et al.	N/A	N/A
2006/0053174	12/2005	Gardner et al.	N/A	N/A
2006/0074621	12/2005	Rachman	N/A	N/A
2006/0112123	12/2005	Clark et al.	N/A	N/A
2006/0129554	12/2005	Suyama et al.	N/A	N/A
2006/0136194	12/2005	Armstrong et al.	N/A	N/A
2006/0143159	12/2005	Chowdhury et al.	N/A	N/A
2006/0173917	12/2005	Kalmick et al.	N/A	N/A
2006/0190804	12/2005	Yang	N/A	N/A
2006/0225001	12/2005	Sylthe et al.	N/A	N/A
2006/0253423	12/2005	McLane et al.	N/A	N/A
2006/0253790	12/2005	Ramarajan et al.	N/A	N/A
2006/0265397	12/2005	Bryan et al.	N/A	N/A
2006/0271520	12/2005	Ragan	N/A	N/A
2006/0277482	12/2005	Hoffman et al.	N/A	N/A
2006/0293979	12/2005	Cash et al.	N/A	N/A
2007/0043703	12/2006	Bhattacharya et al.	N/A	N/A
2007/0061735	12/2006	Hoffberg et al.	N/A	N/A
2007/0061751	12/2006	Cory et al.	N/A	N/A
2007/0100834	12/2006	Landry et al.	N/A	N/A
2007/0118491	12/2006	Baum et al.	N/A	N/A
2007/0209080	12/2006	Ture et al.	N/A	N/A
2007/0214134	12/2006	Haselden et al.	N/A	N/A
2007/0214164	12/2006	MacLennan et al.	N/A	N/A
2007/0239694	12/2006	Singh et al.	N/A	N/A
2008/0021748	12/2007	Bay et al.	N/A	N/A
2008/0097789	12/2007	Huffer	N/A	N/A
2008/0104046	12/2007	Singla et al.	N/A	N/A
2008/0104542	12/2007	Cohen et al.	N/A	N/A
2008/0134071	12/2007	Keohane et al.	N/A	N/A
2008/0177689	12/2007	Jeng et al.	N/A	N/A
2008/0208820	12/2007	Usey et al.	N/A	N/A
2008/0215546	12/2007	Baum et al.	N/A	N/A
2008/0222125	12/2007	Chowdhury	N/A	N/A
2008/0235277	12/2007	Mathew et al.	N/A	N/A
2008/0249858	12/2007	Angell et al.	N/A	N/A
2008/0270366	12/2007	Frank	N/A	N/A
2008/0291030	12/2007	Pape et al.	N/A	N/A
2008/0294740	12/2007	Grabarnik et al.	N/A	N/A
2008/0301095	12/2007	Zhu et al.	N/A	N/A
2008/0306980	12/2007	Brunner et al.	N/A	N/A
2008/0320033	12/2007	Koistinen et al.	N/A	N/A
2009/0055523	12/2008	Song et al.	N/A	N/A
2009/0094207	12/2008	Marvit et al.	N/A	N/A

2009/0125916	12/2008	Lu et al.	N/A	N/A
2009/0177689	12/2008	Song et al.	N/A	N/A
2009/0198640	12/2008	To	N/A	N/A
2009/0216867	12/2008	Pusateri et al.	N/A	N/A
2009/0265424	12/2008	Kimoto et al.	N/A	N/A
2009/0300065	12/2008	Birchall	N/A	N/A
2009/0319512	12/2008	Baker et al.	N/A	N/A
2009/0319941	12/2008	Laansoo et al.	N/A	N/A
2009/0327319	12/2008	Bertram et al.	N/A	N/A
2010/0015211	12/2009	Barnett et al.	N/A	N/A
2010/0017390	12/2009	Yamasaki et al.	N/A	N/A
2010/0057660	12/2009	Kato	N/A	N/A
2010/0095018	12/2009	Khemani et al.	N/A	N/A
2010/0106743	12/2009	Brunner et al.	N/A	N/A
2010/0138377	12/2009	Wright et al.	N/A	N/A
2010/0223499	12/2009	Panigrahy et al.	N/A	N/A
2010/0229096	12/2009	Maiocco et al.	N/A	N/A
2010/0250236	12/2009	Jagannathan et al.	N/A	N/A
2010/0250497	12/2009	Redlich et al.	N/A	N/A
2010/0251100	12/2009	Delacourt	N/A	N/A
2010/0275128	12/2009	Ward et al.	N/A	N/A
2010/0306281	12/2009	Williamson	N/A	N/A
2010/0333008	12/2009	Taylor	N/A	N/A
2011/0029817	12/2010	Nakagawa et al.	N/A	N/A
2011/0035345	12/2010	Duan et al.	N/A	N/A
2011/0040724	12/2010	Dircz	N/A	N/A
2011/0066585	12/2010	Subrahmanyam et al.	N/A	N/A
2011/0066632	12/2010	Robson et al.	N/A	N/A
2011/0119219	12/2010	Naifeh et al.	N/A	N/A
2011/0137836	12/2010	Kuriyama et al.	N/A	N/A
2011/0153646	12/2010	Hong et al.	N/A	N/A
2011/0219035	12/2010	Korsunsky et al.	N/A	N/A
2011/0231223	12/2010	Winters	N/A	N/A
2011/0246528	12/2010	Hsieh et al.	N/A	N/A
2011/0246644	12/2010	Hamada	N/A	N/A
2011/0270877	12/2010	Kim	N/A	N/A
2011/0276695	12/2010	Maldaner	N/A	N/A
2011/0295871	12/2010	Folting et al.	N/A	N/A
2011/0313844	12/2010	Chandramouli et al.	N/A	N/A
2011/0314148	12/2010	Petersen et al.	N/A	N/A
2011/0320450	12/2010	Liu et al.	N/A	N/A
2012/0023429	12/2011	Medhi	N/A	N/A
2012/0054675	12/2011	Rajamannar et al.	N/A	N/A
2012/0079363	12/2011	Folting et al.	N/A	N/A
2012/0094694	12/2011	Malkin et al.	N/A	N/A
2012/0101975	12/2011	Khosravy	N/A	N/A
2012/0131185	12/2011	Petersen et al.	N/A	N/A
2012/0137367	12/2011	Dupont et al.	N/A	N/A
2012/0197914	12/2011	Harnett et al.	N/A	N/A
2012/0221553	12/2011	Wittmer et al.	N/A	N/A

2012/0221580	12/2011	Barney	N/A	N/A
2012/0227004	12/2011	Madireddi et al.	N/A	N/A
2012/0246303	12/2011	Petersen et al.	N/A	N/A
2012/0283948	12/2011	Demiryurek et al.	N/A	N/A
2012/0296876	12/2011	Bacinschi et al.	N/A	N/A
2012/0324329	12/2011	Ceponkus et al.	N/A	N/A
2012/0324359	12/2011	Lee et al.	N/A	N/A
2013/0007645	12/2012	Kurniawan et al.	N/A	N/A
2013/0019019	12/2012	Lam	N/A	N/A
2013/0035961	12/2012	Yegnanarayanan	N/A	N/A
2013/0041824	12/2012	Gupta	N/A	N/A
2013/0054642	12/2012	Morin	N/A	N/A
2013/0054660	12/2012	Zhang	N/A	N/A
2013/0060912	12/2012	Rensin et al.	N/A	N/A
2013/0060937	12/2012	Bath et al.	N/A	N/A
2013/0073542	12/2012	Zhang et al.	N/A	N/A
2013/0073573	12/2012	Huang et al.	N/A	N/A
2013/0073957	12/2012	DiGiantomasso et al.	N/A	N/A
2013/0080190	12/2012	Mansour et al.	N/A	N/A
2013/0080641	12/2012	Lui et al.	N/A	N/A
2013/0103409	12/2012	Malkin et al.	N/A	N/A
2013/0144863	12/2012	Mayer et al.	N/A	N/A
2013/0173322	12/2012	Gray	N/A	N/A
2013/0182700	12/2012	Figura et al.	N/A	N/A
2013/0185643	12/2012	Greifeneder et al.	N/A	N/A
2013/0205014	12/2012	Muro et al.	N/A	N/A
2013/0232094	12/2012	Anderson et al.	N/A	N/A
2013/0238631	12/2012	Carmel et al.	N/A	N/A
2013/0262371	12/2012	Nolan	N/A	N/A
2013/0318236	12/2012	Coates et al.	N/A	N/A
2014/0019909	12/2013	Leonard et al.	N/A	N/A
2014/0046976	12/2013	Zhang et al.	N/A	N/A
2014/0082513	12/2013	Mills et al.	N/A	N/A
2014/0129942	12/2013	Rathod	N/A	N/A
2014/0160238	12/2013	Mm et al.	N/A	N/A
2014/0324862	12/2013	Bingham et al.	N/A	N/A
2015/0019537	12/2014	Neels et al.	N/A	N/A
2015/0058318	12/2014	Blackwell et al.	N/A	N/A
2015/0109305	12/2014	Black	N/A	N/A
2015/0142725	12/2014	Candea et al.	N/A	N/A
2015/0213631	12/2014	Vander Broek	N/A	N/A
2015/0222604	12/2014	Ylonen	N/A	N/A
2015/0278153	12/2014	Leonard et al.	N/A	N/A
2015/0278214	12/2014	Anand et al.	N/A	N/A
2015/0294256	12/2014	Mahesh et al.	N/A	N/A
2015/0341212	12/2014	Hsiao et al.	N/A	N/A
2016/0092045	12/2015	Lamas et al.	N/A	N/A
2016/0224531	12/2015	Robichaud et al.	N/A	N/A
2016/0224614	12/2015	Robichaud et al.	N/A	N/A
2016/0224618	12/2015	Robichaud et al.	N/A	N/A

2016/0224619	12/2015	Robichaud et al.	N/A	N/A
2016/0224624	12/2015	Robichaud	N/A	N/A
2016/0224643	12/2015	Robichaud	N/A	N/A
2016/0224659	12/2015	Robichaud	N/A	N/A
2016/0224804	12/2015	Carasso	N/A	N/A
2016/0350950	12/2015	Ritchie et al.	N/A	N/A
2017/0011229	12/2016	Jones-McFadden et al.	N/A	N/A
2017/0139996	12/2016	Marquardt et al.	N/A	N/A
2017/0286038	12/2016	Li et al.	N/A	N/A
2017/0286455	12/2016	Li et al.	N/A	N/A
2017/0286525	12/2016	Li et al.	N/A	N/A
2017/0322959	12/2016	Tidwell et al.	N/A	N/A
2018/0032706	12/2017	Fox et al.	N/A	N/A
2018/0089303	12/2017	Miller et al.	N/A	N/A
2018/0089561	12/2017	Oliner et al.	N/A	N/A
2018/0314853	12/2017	Oliner et al.	N/A	N/A
2019/0098106	12/2018	Mungel et al.	N/A	N/A
2019/0146978	12/2018	Beedgen et al.	N/A	N/A
2019/0268354	12/2018	Zettel, II et al.	N/A	N/A
2020/0125725	12/2019	Petersen et al.	N/A	N/A
2020/0135337	12/2019	Athey et al.	N/A	N/A
2020/0143054	12/2019	Cohen et al.	N/A	N/A
2020/0336505	12/2019	Neuvirth et al.	N/A	N/A
2021/0117232	12/2020	Sriharsha et al.	N/A	N/A
2021/0117425	12/2020	Rao et al.	N/A	N/A
2021/0117868	12/2020	Sriharsha	N/A	N/A
2021/0133634	12/2020	Ma et al.	N/A	N/A
2021/0374112	12/2020	Nagai et al.	N/A	N/A
2022/0121708	12/2021	Burnett et al.	N/A	N/A
2022/0208339	12/2021	Neumann	N/A	N/A
2022/0245093	12/2021	Batsakis et al.	N/A	N/A
2022/0269727	12/2021	Batsakis et al.	N/A	N/A
2023/0237049	12/2022	Singh et al.	N/A	N/A
2024/0143612	12/2023	Bigdelu	N/A	G06F 16/2425
2025/0036645	12/2024	Bigdelu	N/A	G06F 16/2425

OTHER PUBLICATIONS

U.S. Appl. No. 17/733,617, filed Apr. 29, 2022, Patented Case. cited by applicant
U.S. Appl. No. 17/384,467, filed Jul. 23, 2021, Patented Case. cited by applicant
U.S. Appl. No. 13/748,313, filed Jan. 23, 2013, Pending. cited by applicant
U.S. Appl. No. 14/611,093, filed Jan. 30, 2015, Abandoned. cited by applicant
U.S. Appl. No. 14/169,268, filed Jan. 31, 2014, Abandoned. cited by applicant
U.S. Appl. No. 15/582,671, filed Apr. 29, 2017, Pending. cited by applicant
U.S. Appl. No. 15/582,670, filed Apr. 29, 2017, Patented Case. cited by applicant
U.S. Appl. No. 17/028,722, filed Sep. 22, 2020, Pending. cited by applicant
U.S. Appl. No. 14/610,668, filed Jan. 30, 2015, Pending. cited by applicant
U.S. Appl. No. 14/611,089, filed Jan. 30, 2015, Pending. cited by applicant
U.S. Appl. No. 15/417,430, filed Jan. 27, 2017, Pending. cited by applicant
U.S. Appl. No. 13/747,177, filed Jan. 22, 2013, Abandoned. cited by applicant
U.S. Appl. No. 15/582,599, filed Apr. 28, 2017, Pending. cited by applicant

U.S. Appl. No. 15/694,654, filed Sep. 1, 2017, Pending. cited by applicant
U.S. Appl. No. 16/394,754, filed Apr. 25, 2019, Pending. cited by applicant
U.S. Appl. No. 16/541,637, filed Aug. 15, 2019, Pending. cited by applicant
U.S. Appl. No. 16/587,445, filed Oct. 1, 2019, Pending. cited by applicant
U.S. Appl. No. 16/003,998, filed Jun. 8, 2018, Pending. cited by applicant
U.S. Appl. No. 17/028,755, filed Sep. 22, 2020, Pending. cited by applicant
U.S. Appl. No. 14/266,839, filed May 1, 2014, Pending. cited by applicant
U.S. Appl. No. 13/748,391, filed Jan. 23, 2013, Pending. cited by applicant
U.S. Appl. No. 13/607,117, filed Sep. 7, 2012, Pending. cited by applicant
U.S. Appl. No. 14/067,203, filed Oct. 30, 2013, Pending. cited by applicant
U.S. Appl. No. 13/747,153, filed Jan. 22, 2013, Pending. cited by applicant
U.S. Appl. No. 14/168,888, filed Jan. 30, 2014, Pending. cited by applicant
U.S. Appl. No. 13/748,360, filed Jan. 23, 2013, Pending. cited by applicant
U.S. Appl. No. 14/816,036, filed Aug. 2, 2015, Pending. cited by applicant
U.S. Appl. No. 14/816,038, filed Aug. 2, 2015, Pending. cited by applicant
U.S. Appl. No. 15/011,392, filed Jan. 29, 2016, Abandoned. cited by applicant
U.S. Appl. No. 15/582,667, filed Apr. 29, 2017, Pending. cited by applicant
U.S. Appl. No. 15/582,668, filed Apr. 29, 2017, Pending. cited by applicant
U.S. Appl. No. 15/582,669, filed Apr. 29, 2017, Patented Case. cited by applicant
U.S. Appl. No. 17/169,254, filed Feb. 5, 2021, Pending. cited by applicant
U.S. Appl. No. 17/874,046, filed Jul. 26, 2022, Patented Case. cited by applicant
U.S. Appl. No. 17/443,892, filed Jul. 28, 2021, Patented Case. cited by applicant
U.S. Appl. No. 18/306,863, filed Apr. 25, 2023, Pending. cited by applicant
U.S. Appl. No. 17/968,691, filed Oct. 18, 2022, Pending. cited by applicant
Aho, A.V., et al., “Columbia digital news project: an environment for briefing and search over multimedia information,” International Journal on Digital Libraries, vol. 1, No. 4, pp. 377-385 (Mar. 1998). cited by applicant
Carasso, D., “Exploring Splunk: Search Processing Language (SPL) Primer and Coockbook,” Splunk, pp. 156 (Apr. 2012). cited by applicant
Carasso, D., “Field Extractor App (Walkthrough),” [online video excerpts], YouTube, Jul. 12, 2013, Retrieved from the Internet: <https://www.youtube.com/walch?v=Gfl9Cm9v64Y> on Jun. 17, 2014, last accessed on May 19, 2015, pp. 2. cited by applicant
Carasso, D., “Semi-Automatic Discovery of Extraction Patterns for Log Analysis,” Splunk Inc., pp. 3 (2007). cited by applicant
Ennis, Mark; “Headache relief for programmers:: regular expression generator,” Aug. 13, 2007, http://txt2re.com/, accessed on Jan. 15, 2016, p. 1. cited by applicant
“iTunes 10 for Mac: Create a Smart Playlist”, Apple, Nov. 27, 2012, accessed at http://support.apple.com/kb/PH1739?viewlocale=en_US, accessed on Feb. 18, 2013, p. 1. cited by applicant
Patel, D., et al., “An evaluation of techniques for image searching and browsing on mobile devices,” Published in: Proceeding SAICSIT '09 Proceedings of the 2009 Annual Research Conference of the South African Institute of Computer Scientists and Information Technologies, pp. 60-69 (Oct. 2009). cited by applicant
Tong, H.H., et al., “Fast mining of complex time-stamped events,” Proceeding CIKM '08 Proceedings of the 17th ACM conference on Information and knowledge management, pp. 759-767 (Oct. 26-30, 2008). cited by applicant
Kalmanek, C.R., et al., “Darkstar: Using exploratory data mining to raise the bar on network reliability and performance,” 2009 7th International Workshop on Design of Reliable Communication Networks, pp. 1-10 (2009). cited by applicant
Wang, M., et al., “Event Indexing and Searching for High vols. of Event Streams in the Cloud,”

2012 IEEE 36th Annual Date of Conference Published in Computer Software and Applications Conference (COMPSAC), pp. 405-415 (Jul. 16-20, 2012). cited by applicant
"RegexBuddy Demo-Self-Running Demonstration," RegexBuddy.com, Oct. 28, 2012, accessed at http://www.regexbuddy.com/democreate.html, accessed on Feb. 18, 2013, pp. 2. cited by applicant
Riloff, E., et al., "Learning Dictionaries for information Extraction by Multi-Level Bootstrapping," Proceedings of the Sixteenth National Conference on Artificial Intelligence, pp. 6 (Jul. 1999). cited by applicant
Soderland, S., et al., "Issues in Inductive Learning of Domain-Specific Text Extraction Rules," Proceedings of the Workshop on New Approaches to Learning for Natural language Processing at the Fourteenth International Joint Conference on Artificial Intelligence, pp. 1-4 (Aug. 1995). cited by applicant
Splunk, Splunk User Manual Version 4.1, pp. 181 (Year: 2011). cited by applicant
Txt2re.com Google Search, hllps://www.google.com/search?q=txt2re.com&biw=1536&bih=824&source=Int&tbs=cdr . . . , accessed on Jan. 19, 2016, pp. 2. cited by applicant

---

Primary Examiner: Phantana-Angkool; David

Attorney, Agent or Firm: Shook, Hardy & Bacon L.L.P

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/384,467, filed Jul. 23, 2021, entitled "Graphical User Interface for Extraction Rules", which is itself a continuation of U.S. application Ser. No. 17/028,755, filed Sep. 22, 2020, entitled "Displaying Event Records with Emphasized Fields", which is a continuation of U.S. application Ser. No. 16/003,998, filed Jun. 8, 2018, entitled "Providing An Extraction Rule Associated With A Selected Portion Of An Event," which is a continuation of U.S. application Ser. No. 14/266,839, filed May 1, 2014, entitled "Automatic Extraction Rule Generation Based On User Selections Of Text Value," which is a continuation of U.S. application Ser. No. 13/748,391, filed Jan. 23, 2013, entitled "Real Time Indication Of Previously Extracted Data Fields For Regular Expressions." The entire contents of each of the foregoing applications are herein incorporated by reference.

TECHNICAL FIELD
(1) The present invention relates generally to data presentation management, and more particularly, but not exclusively, to providing real time display of event records with an indication of fields defined by previously provided extraction rules.
BRIEF SUMMARY OF THE INVENTION
(2) The rapid increase in the production and collection of machine-generated data has created large data sets that are difficult to search and/or otherwise analyze. The machine data can include sequences of time stamped records that may occur in one or more usually continuous streams. Further, machine data often represents activity made up of discrete records or events.
(3) Often, search engines may receive data from various data sources, including machine data. In some cases, this data may be analyzed or processed in a variety of ways. However, prior to such processing field values may need to be extracted from the received data. Sometimes the received data may be unstructured, which may make it difficult for systems to efficiently analyze the received data to determine what data may be of interest and/or how to generate a field value extraction rule. This may be especially true where the datasets are considered extremely large, such

as terabytes or greater. Such large unstructured datasets may make it difficult and time consuming to analyze the data so as to be able to perform various actions on the data. For example, determining extraction rules, modification rules, or the like on such large datasets that are correct and effective may be difficult and time consuming. Improper and/or ineffective rules may result in improper value from the received data and/or omit significant values. Thus, it is with respect to these considerations and others that the present invention has been made.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Non-limiting and non-exhaustive embodiments are described with reference to the following drawings. In the drawings, like reference numerals refer to like parts throughout the various figures unless otherwise specified.

(2) For a better understanding, reference will be made to the following Detailed Description, which is to be read in association with the accompanying drawings, wherein:

(3) FIG. **1** illustrates a system environment in which various embodiments may be implemented;

(4) FIG. **2**A shows a rack of blade servers that may be included in various embodiments;

(5) FIG. **2**B illustrates an embodiment of a blade server that may be included in a rack of blade servers such as that shown in FIG. **2**A;

(6) FIG. **3** shows a client device that may be included in various embodiments;

(7) FIG. **4** illustrates a network device that may be included in various embodiments;

(8) FIG. **5** illustrates a logical flow diagram generally showing one embodiment of an overview process for enabling real time display of fields based on previously provided extraction rules;

(9) FIG. **6** illustrates a logical flow diagram generally showing one embodiment of a process for displaying event records that emphasizes fields based on previously provided extraction rules;

(10) FIG. **7** illustrates a logical flow diagram generally showing one embodiment of a process for displaying previously provided extraction rules associated with a selected portion of an event record;

(11) FIGS. **8**A-**8**C illustrate non-exhaustive examples of a use case of embodiments of a graphical user interface that may be employed to enable a user to create extraction rules and to display indications of previously extracted data fields;

(12) FIGS. **9**A-**9**B illustrate non-exhaustive examples of a use case of embodiments of a graphical user interface, such as depicted in FIGS. **8**A-**8**C, to display extraction rules and/or fields associated with a selected portion of event data; and

(13) FIGS. **10**A-**10**C illustrate a non-exhaustive example of a code fragment for determining a part of an event record to emphasize based on a plurality of fields.

DETAILED DESCRIPTION OF THE INVENTION

(14) Various embodiments now will be described more fully hereinafter with reference to the accompanying drawings, which form a part hereof, and which show, by way of illustration, specific embodiments by which the invention may be practiced. The embodiments may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the embodiments to those skilled in the art. Among other things, the various embodiments may be methods, systems, media, or devices. Accordingly, the various embodiments may take the form of an entirely hardware embodiment, an entirely software embodiment, or an embodiment combining software and hardware aspects. The following detailed description is, therefore, not to be taken in a limiting sense.

(15) Throughout the specification and claims, the following terms take the meanings explicitly associated herein, unless the context clearly dictates otherwise. The phrase “in one embodiment” as

used herein does not necessarily refer to the same embodiment, though it may. Furthermore, the phrase "in another embodiment" as used herein does not necessarily refer to a different embodiment, although it may. Thus, as described below, various embodiments may be readily combined, without departing from the scope or spirit of the invention.

(16) In addition, as used herein, the term "or" is an inclusive "or" operator, and is equivalent to the term "and/or," unless the context clearly dictates otherwise. The term "based on" is not exclusive and allows for being based on additional factors not described, unless the context clearly dictates otherwise. In addition, throughout the specification, the meaning of "a," "an," and "the" include plural references. The meaning of "in" includes "in" and "on."

(17) For example embodiments, the following terms are also used herein according to the corresponding meaning, unless the context clearly dictates otherwise.

(18) The term "machine data" as used herein may include data generated by machines, including, but not limited to, server logs or other types of event data and/or event records. In at least one of various embodiments, machine data streams may be time stamped to create time stamped events. For example, information processing environments, such as, firewalls, routers, web servers, application servers and databases may generate streams of time series data in the form of events. In some cases, events may be generated hundreds or thousands of times per second. In some embodiments, the machine data may be unstructured data, structured data, and/or a combination thereof. Unstructured data may refer to data that does not include at least one predefined field.

(19) As used herein, the term "event record" may refer to computing data that is collected about an event for a computing system, including, for example, an action, characteristic, condition (or state) of the computing system. For example, such events may be about a computing system's performance, actions taken by the computing system, or the like. Event records may be obtained from various computing log files generated by the computer's operating system, and/or other monitoring application. However, event records are not restricted by a file format or structure from which the event data is obtained. In various embodiments, event records may include structured and/or unstructured machine data, and/or any combination thereof.

(20) The term "extraction rule" and/or "data field extraction rule" may refer to instructions that may be applied to identify and extract field values from data, such as event records. In some embodiments, extraction rule may define a field within event records from which to extract a value. In at least one of various embodiments, the extraction rules may include regular expressions. The data from which extraction rules may be applied may include structured and/or unstructured machine data, or other type of data.

(21) The term "regular expression" as used herein may refer to a sequence of constants and operators arranged into expressions for matching a set of strings. A regular expression is often defined as a pattern matching language which can be employed to identify character strings, for example, to select specific strings from a set of character strings. More particularly, regular expressions are often defined as a context-independent syntax that can represent a wide variety of character sets and character set orderings. In operation, regular expressions can be employed to search data based upon a predefined pattern or set of patterns. As such, this pattern matching language employs a specific syntax by which particular characters or strings are selected from a body of text. Although simple examples of regular expressions can be easily understood, oftentimes, the syntax of regular expressions are so complex that even the most experienced programmers have difficulty in understanding them. Regular expressions may be constructed using a variety of computer languages and constructs. In addition to matching, some regular expression systems offer functionality, such as, substitution, grouping, back references, or the like. Regular expressions and regular expression systems may be adapted to work with non-string data providing matching facilities for binary data.

(22) The following briefly describes embodiments in order to provide a basic understanding of some aspects of the invention. This brief description is not intended as an extensive overview. It is

not intended to identify key or critical elements, or to delineate or otherwise narrow the scope. Its purpose is merely to present some concepts in a simplified form as a prelude to the more detailed description that is presented later.

(23) Briefly stated, various embodiments are directed towards real time display of event records with an indication of previously provided extraction rules. In at least one of various embodiments, a plurality of extraction rules may be provided to the system. In at least one embodiment, a user interface may be employed to enable a user to have an extraction rule automatically generated and/or to manually enter an extraction rule. In at least one embodiment, the extraction rules may include regular expressions. A plurality of event records may be displayed to the user, such that text in a field defined by at least one of the extraction rules is emphasized in a display of the event record. In at least one of various embodiments, emphasizing the display of text in the field may include one or more updates to the displayed text, such as dimming, highlighting, underlining, balding, striking through, italicizing, displaying different font, displaying different font size, displaying different color, displaying different transparency, including parenthesis around the text, and the like. In some embodiments, the same emphasis may be provided for text in overlapping fields, and in other embodiments it may be at least somewhat different for text in overlapping fields.

(24) The user interface may also be employed to enable the user to select at least a portion of text in at least one of the event records. Once selected, the interface may display each extraction rule associated with the selected portion. In some embodiments, the user may roll-over or click on an emphasized part of the event record to select text in an event record, which may display each extraction rule associated with the selected text of the event record.

(25) Illustrative Operating Environment

(26) FIG. **1** shows components of an environment in which various embodiments may be practiced. Not all of the components may be required to practice the various embodiments, and variations in the arrangement and type of the components may be made without departing from the spirit or scope of the various embodiments.

(27) In at least one embodiment, cloud network **102** enables one or more network services for a user based on the operation of corresponding arrangements **104** and **106** of virtually any type of networked computing device. As shown, the networked computing devices may include extraction rule server device **112**, event records server device **114**, enclosure of blade servers **110**, enclosure of server computers **116**, super computer network device **118**, and the like. Although not shown, one or more client devices may be included in cloud network **102** in one or more arrangements to provide one or more network services to a user. Also, these arrangements of networked computing devices may or may not be mutually exclusive of each other.

(28) Additionally, the user may employ a plurality of virtually any type of wired or wireless networked computing devices to communicate with cloud network **102** and access at least one of the network services enabled by one or more of arrangements **104** and **106**. These networked computing devices may include tablet client device **122**, handheld client device **124**, wearable client device **126**, desktop client device **120**, and the like. Although not shown, in various embodiments, the user may also employ notebook computers, desktop computers, microprocessor-based or programmable consumer electronics, network appliances, mobile telephones, smart telephones, pagers, radio frequency (RF) devices, infrared (IR) devices, Personal Digital Assistants (PDAs), televisions, integrated devices combining at least one of the preceding devices, and the like.

(29) One embodiment of a client device is described in more detail below in conjunction with FIG. **3**. Generally, client devices may include virtually any substantially portable networked computing device capable of communicating over a wired, wireless, or some combination of wired and wireless network.

(30) In various embodiments, network **102** may employ virtually any form of communication

technology and topology. For example, network **102** can include local area networks Personal Area Networks (PANs), (LANs), Campus Area Networks (CANs), Metropolitan Area Networks (MANs) Wide Area Networks (WANs), direct communication connections, and the like, or any combination thereof. On an interconnected set of LANs, including those based on differing architectures and protocols, a router acts as a link between LANs, enabling messages to be sent from one to another. In addition, communication links within networks may include virtually any type of link, e.g., twisted wire pair lines, optical fibers, open air lasers or coaxial cable, plain old telephone service (POTS), wave guides, acoustic, full or fractional dedicated digital communication lines including T1, T2, T3, and T4, and/or other carrier and other wired media and wireless media. These carrier mechanisms may include E-carriers, Integrated Services Digital Networks (ISDNs), universal serial bus (USB) ports, Firewire ports, Thunderbolt ports, Digital Subscriber Lines (DSLs), wireless links including satellite links, or other communications links known to those skilled in the art. Moreover, these communication links may further employ any of a variety of digital signaling technologies, including without limit, for example, DS-0, DS-1, DS-2, DS-3, DS-4, OC-3, OC-12, OC-48, or the like. Furthermore, remotely located computing devices could be remotely connected to networks via a modem and a temporary communication link. In essence, network **102** may include virtually any communication technology by which information may travel between computing devices. Additionally, in the various embodiments, the communicated information may include virtually any kind of information including, but not limited to processor-readable instructions, data structures, program modules, applications, raw data, control data, archived data, video data, voice data, image data, text data, and the like.

(31) Network **102** may be partially or entirely embodied by one or more wireless networks. A wireless network may include any of a variety of wireless sub-networks that may further overlay stand-alone ad-hoc networks, and the like. Such sub-networks may include mesh networks, Wireless LAN (WLAN) networks, Wireless Router (WR) mesh, cellular networks, pico networks, PANs, Open Air Laser networks, Microwave networks, and the like. Network **102** may further include an autonomous system of intermediate network devices such as terminals, gateways, routers, switches, firewalls, load balancers, and the like, which are coupled to wired and/or wireless communication links. These autonomous devices may be operable to move freely and randomly and organize themselves arbitrarily, such that the topology of network **102** may change rapidly.

(32) Network **102** may further employ a plurality of wired and wireless access technologies, e.g., 2nd (20), 3rd (30), 4th (40), 5th (50) generation wireless access technologies, and the like, for mobile devices. These wired and wireless access technologies may also include Global System for Mobile communication (GSM), General Packet Radio Services (GPRS), Enhanced Data GSM Environment (EDGE), Code Division Multiple Access (CDMA), Wideband Code Division Multiple Access (WCDMA), Long Term Evolution Advanced (LTE), Universal Mobile Telecommunications System (UMTS), Orthogonal frequency-division multiplexing (OFDM), Wideband Code Division Multiple Access (W-CDMA), Code Division Multiple Access 2000 (CDMA2000), Evolution-Data Optimized (EV-DO), High-Speed Downlink Packet Access (HSDPA), IEEE 802.16 Worldwide Interoperability for Microwave Access (WiMax), ultra wide band (UWB), user datagram protocol (UDP), transmission control protocol/Internet protocol (TCP/IP), any portion of the Open Systems Interconnection (OSI) model protocols, Short Message Service (SMS), Multimedia Messaging Service (MMS), Web Access Protocol (W AP), Session Initiation Protocol/Real-time Transport Protocol (SIP/RTP), or any of a variety of other wireless or wired communication protocols. In one non-limiting example, network **102** may enable a mobile device to wirelessly access a network service through a combination of several radio network access technologies such as GSM, EDGE, SMS, HSDPA, and the like.

(33) One embodiment of extraction rule server device **112** is described in more detail below in conjunction with FIG. **4**. Briefly, however, extraction rule server device **112** includes virtually any network device capable of generating and/or managing extraction rules. In some embodiments,

extraction rule server device **112** may automatically generate extraction rules. Devices that may be arranged to operate as extraction rule server device **112** include various network devices, including, but not limited to personal computers, desktop computers, multiprocessor systems, microprocessor-based or programmable consumer electronics, network PCs, server devices, network appliances, and the like.

(34) Although FIG. **1** illustrates extraction rule server device **112** as a single computing device, the invention is not so limited. For example, one or more functions of the extraction rule server device **112** may be distributed across one or more distinct network devices. Moreover, extraction rule server device **112** is not limited to a particular configuration. Thus, in one embodiment, extraction rule server device **112** may contain a plurality of network devices. In another embodiment, extraction rule server device **112** may contain a plurality of network devices that operate using a master/slave approach, where one of the plurality of network devices of extraction rule server device **112** operates to manage and/or otherwise coordinate operations of the other network devices. In other embodiments, the extraction rule server device **112** may operate as a plurality of network devices within a cluster architecture, a peer-to-peer architecture, and/or even within a cloud architecture. Thus, the invention is not to be construed as being limited to a single environment, and other configurations, and architectures are also envisaged.

(35) One embodiment of event records server device **114** is described in more detail below in conjunction with FIG. **4**. Briefly, however, event records server device **114** includes virtually any network device capable of collecting and/or maintaining event records. In some embodiments, event records server device **114** may be in communication with extraction rule server device **112** to enable the use of extraction rules on event records. Devices that may be arranged to operate as event records server device **114** include various network devices, including, but not limited to personal computers, desktop computers, multiprocessor systems, microprocessor-based or programmable consumer electronics, network PCs, server devices, network appliances, and the like.

(36) Although FIG. **1** illustrates event records server device **114** as a single computing device, the invention is not so limited. For example, one or more functions of the event records server device **114** may be distributed across one or more distinct network devices. Moreover, event records server device **114** is not limited to a particular configuration. Thus, in one embodiment, event records server device **114** may contain a plurality of network devices. In another embodiment, event records server device **114** may contain a plurality of network devices that operate using a master/slave approach, where one of the plurality of network devices of event records server device **114** operates to manage and/or otherwise coordinate operations of the other network devices. In other embodiments, the event records server device **114** may operate as a plurality of network devices within a cluster architecture, a peer-to-peer architecture, and/or even within a cloud architecture. Thus, the invention is not to be construed as being limited to a single environment, and other configurations, and architectures are also envisaged.

(37) Enclosure of Blade Servers

(38) FIG. **2**A shows one embodiment of an enclosure of blade servers **200**, which are also illustrated in FIG. **1**. Enclosure of blade servers **200** may include many more or fewer components than those shown in FIG. **2**A. However, the components shown are sufficient to disclose an illustrative embodiment. Generally, a blade server is a stripped down server computing device with a modular design optimized to minimize the use of physical space and energy. A blade enclosure can include several blade servers and provide each with power, cooling, network interfaces, input/output interfaces, and resource management. Although not shown, an enclosure of server computers typically includes several computers that merely require a network connection and a power cord connection to operate. Each server computer often includes redundant components for power and interfaces.

(39) As shown in the figure, enclosure **200** contains power supply **204**, and input/output interface

206, rack logic **208**, several blade servers **210**, **212**, **214**, and **216**, and backplane **202**. Power supply **204** provides power to each component and blade server within the enclosure. The input/output interface **206** provides internal and external communication for components and blade servers within the enclosure. Backplane **208** can enable passive and active communication of power, logic, input signals, and output signals for each blade server.

(40) Illustrative Blade Server

(41) FIG. **2**B illustrates an illustrative embodiment of blade server **250**, which may include many more or fewer components than those shown. As shown in FIG. **2**A, a plurality of blade servers may be included in one enclosure that shares resources provided by the enclosure to reduce size, power, and cost.

(42) Blade server **250** may include processor **252** which communicates with memory **256** via bus **254**. Blade server **250** may also include input/output interface **290**, processor-readable stationary storage device **292**, and processor-readable removable storage device **294**. Input/output interface **290** can enable blade server **250** to communicate with other blade servers, client devices, network devices, and the like. Interface **290** may provide wireless and/or wired communication links for blade server. Processor-readable stationary storage device **292** may include devices such as an electromagnetic storage device (hard disk), solid state hard disk (SSD), hybrid of both an SSD and a hard disk, and the like. Also, processor-readable removable storage device **294** enables processor **252** to read non-transitive storage media for storing and accessing processor-readable instructions, modules, data structures, and other forms of data. The non-transitive storage media may include Flash drives, tape media, floppy media, and the like.

(43) Memory **256** may include Random Access Memory (RAM), Read-Only Memory (ROM), hybrid of RAM and ROM, and the like. As shown, memory **256** includes operating system **258** and basic input/output system (BIOS) **260** for enabling the operation of blade server **250**. In various embodiments, a general-purpose operating system may be employed such as a version of UNIX, or LINUX™, or a specialized server operating system such as Microsoft's Windows Server™ and Apple Computer's iOS Server™.

(44) Memory **256** may further include one or more data storage **270**, which can be utilized by blade server **250** to store, among other things, applications **280** and/or other data. Data stores **270** may include program code, data, algorithms, and the like, for use by processor **252** to execute and perform actions. In one embodiment, at least some of data store **270** might also be stored on another component of blade server **250**, including, but not limited to, processor-readable removable storage device **294**, processor-readable stationary storage device **292**, or any other processo-rreadable storage device (not shown). Data storage **270** may include, for example, event records **272** and extraction rules **274**.

(45) Applications **280** may include processor executable instructions which, when executed by blade server **250**, transmit, receive, and/or otherwise process messages, audio, video, and enable communication with other networked computing devices. Examples of application programs include database servers, file servers, calendars, transcoders, and so forth. Applications **280** may include, for example, extraction rule application **282**.

(46) Human interface components (not pictured), may be remotely associated with blade server **250**, which can enable remote input to and/or output from blade server **250**. For example, information to a display or from a keyboard can be routed through the input/output interface **290** to appropriate peripheral human interface components that are remotely located. Examples of peripheral human interface components include, but are not limited to, an audio interface, a display, keypad, pointing device, touch interface, and the like.

(47) Illustrative Client Device

(48) FIG. **3** shows one embodiment of client device **300** that may include many more or less components than those shown. Client device **300** may represent, for example, at least one embodiment of client devices shown in FIG. **1**.

(49) Client device **300** may include processor **302** in communication with memory **304** via bus **328**. Client device **300** may also include power supply **330**, network interface **332**, audio interface **356**, display **350**, keypad **352**, illuminator **354**, video interface **342**, input/output interface **338**, haptic interface **364**, global positioning systems (GPS) receiver **358**, open air gesture interface **360**, temperature interface **362**, camera(s) **340**, projector **346**, pointing device interface **366**, processor-readable stationary storage device **334**, and processor-readable removable storage device **336**. Client device **300** may optionally communicate with a base station (not shown), or directly with another computing device. And in one embodiment, although not shown, a gyroscope may be employed within client device **300** to measuring and/or maintaining an orientation of client device **300**.
(50) Power supply **330** may provide power to client device **300**. A rechargeable or non-rechargeable battery may be used to provide power. The power may also be provided by an external power source, such as an AC adapter or a powered docking cradle that supplements and/or recharges the battery.
(51) Network interface **332** includes circuitry for coupling client device **300** to one or more networks, and is constructed for use with one or more communication protocols and technologies including, but not limited to, protocols and technologies that implement any portion of the OSI model for mobile communication (GSM), CDMA, time division multiple access (TDMA), UDP, TCP/IP, SMS, MMS, GPRS, WAP, UWB, WiMax, SIP/RTP, GPRS, EDGE, WCDMA, LTE, UMTS, OFDM, CDMA2000, EV-DO, HSDPA, or any of a variety of other wireless communication protocols. Network interface **332** is sometimes known as a transceiver, transceiving device, or network interface card (NIC).
(52) Audio interface **356** may be arranged to produce and receive audio signals such as the sound of a human voice. For example, audio interface **356** may be coupled to a speaker and microphone (not shown) to enable telecommunication with others and/or generate an audio acknowledgement for some action. A microphone in audio interface **356** can also be used for input to or control of client device **300**, e.g., using voice recognition, detecting touch based on sound, and the like.
(53) Display **350** may be a liquid crystal display (LCD), gas plasma, electronic ink, light emitting diode (LED), Organic LED (OLED) or any other type of light reflective or light transmissive display that can be used with a computing device. Display **350** may also include a touch interface **344** arranged to receive input from an object such as a stylus or a digit from a human hand, and may use resistive, capacitive, surface acoustic wave (SAW), infrared, radar, or other technologies to sense touch and/or gestures.
(54) Projector **346** may be a remote handheld projector or an integrated projector that is capable of projecting an image on a remote wall or any other reflective object such as a remote screen.
(55) Video interface **342** may be arranged to capture video images, such as a still photo, a video segment, an infrared video, or the like. For example, video interface **342** may be coupled to a digital video camera, a web-camera, or the like. Video interface **342** may comprise a lens, an image sensor, and other electronics. Image sensors may include a complementary metal-oxide-semiconductor (CMOS) integrated circuit, charge-coupled device (CCD), or any other integrated circuit for sensing light.
(56) Keypad **352** may comprise any input device arranged to receive input from a user. For example, keypad **352** may include a push button numeric dial, or a keyboard. Keypad **352** may also include command buttons that are associated with selecting and sending images.
(57) Illuminator **354** may provide a status indication and/or provide light. Illuminator **354** may remain active for specific periods of time or in response to events. For example, when illuminator **354** is active, it may backlight the buttons on keypad **352** and stay on while the client device is powered. Also, illuminator **354** may backlight these buttons in various patterns when particular actions are performed, such as dialing another client device. Illuminator **354** may also cause light sources positioned within a transparent or translucent case of the client device to illuminate in

response to actions.
(58) Client device **300** may also comprise input/output interface **338** for communicating with external peripheral devices or other computing devices such as other client devices and network devices. The peripheral devices may include an audio headset, display screen glasses, remote speaker system, remote speaker and microphone system, and the like. Input/output interface **338** can utilize one or more technologies, such as Universal Serial Bus (USB), Infrared, WiFi, WiMax, Bluetooth™, and the like.
(59) Haptic interface **364** may be arranged to provide tactile feedback to a user of the client device. For example, the haptic interface **364** may be employed to vibrate client device **300** in a particular way when another user of a computing device is calling. Temperature interface **362** may be used to provide a temperature measurement input and/or a temperature changing output to a user of client device **300**. Open air gesture interface **360** may sense physical gestures of a user of client device **300**, for example, by using single or stereo video cameras, radar, a gyroscopic sensor inside a device held or worn by the user, or the like. Camera **340** may be used to track physical eye movements of a user of client device **300**.
(60) GPS transceiver **358** can determine the physical coordinates of client device **300** on the surface of the Earth, which typically outputs a location as latitude and longitude values. GPS transceiver **358** can also employ other geo-positioning mechanisms, including, but not limited to, triangulation, assisted GPS (AGPS), Enhanced Observed Time Difference (E-OTD), Cell Identifier (CI), Service Area Identifier (SAI), Enhanced Timing Advance (ETA), Base Station Subsystem (BSS), or the like, to further determine the physical location of client device **300** on the surface of the Earth. It is understood that under different conditions, GPS transceiver **358** can determine a physical location for client device **300**. In at least one embodiment, however, client device **300** may, through other components, provide other information that may be employed to determine a physical location of the device, including for example, a Media Access Control (MAC) address, IP address, and the like.
(61) Human interface components can be peripheral devices that are physically separate from client device **300**, allowing for remote input and/or output to client device **300**. For example, information routed as described here through human interface components such as display **350** or keyboard **352** can instead be routed through network interface **332** to appropriate human interface components located remotely. Examples of human interface peripheral components that may be remote include, but are not limited to, audio devices, pointing devices, keypads, displays, cameras, projectors, and the like. These peripheral components may communicate over a Pico Network such as Bluetooth™, Zigbee™ and the like. One non-limiting example of a client device with such peripheral human interface components is a wearable computing device, which might include a remote pico projector along with one or more cameras that remotely communicate with a separately located client device to sense a user's gestures toward portions of an image projected by the pico projector onto a reflected surface such as a wall or the user's hand.
(62) A client device may include a browser application that is configured to receive and to send web pages, web-based messages, graphics, text, multimedia, and the like. The client device's browser application may employ virtually any programming language, including a wireless application protocol messages (W AP), and the like. In at least one embodiment, the browser application is enabled to employ Handheld Device Markup Language (HDML), Wireless Markup Language (WML), WMLScript, JavaScript, Standard Generalized Markup Language (SGML), HyperText Markup Language (HTML), extensible Markup Language (XML), HTML5, and the like.
(63) Memory **304** may include RAM, ROM, and/or other types of memory. Memory **304** illustrates an example of computer-readable storage media (devices) for storage of information such as computer-readable instructions, data structures, program modules or other data. Memory **304** may store BIOS **308** for controlling low-level operation of client device **300**. The memory may also

store operating system **306** for controlling the operation of client device **300**. It will be appreciated that this component may include a general-purpose operating system such as a version of UNIX, or LINUX™, or a specialized mobile computer communication operating system such as Windows Phone™, or the Symbian® operating system. The operating system may include, or interface with a Java virtual machine module that enables control of hardware components and/or operating system operations via Java application programs.

(64) Memory **304** may further include one or more data storage **310**, which can be utilized by client device **300** to store, among other things, applications **320** and/or other data. For example, data storage **310** may also be employed to store information that describes various capabilities of client device **300**. The information may then be provided to another device based on any of a variety of events, including being sent as part of a header during a communication, sent upon request, or the like. Data storage **310** may also be employed to store social networking information including address books, buddy lists, aliases, user profile information, or the like. Data storage **310** may further include program code, data, algorithms, and the like, for use by a processor, such as processor **302** to execute and perform actions. In one embodiment, at least some of data storage **310** might also be stored on another component of client device **300**, including, but not limited to, non-transitory processor-readable removable storage device **336**, processor-readable stationary storage device **334**, or even external to the client device.

(65) Applications **320** may include computer executable instructions which, when executed by client device **300**, transmit, receive, and/or otherwise process instructions and data. Applications **320** may include, for example, extraction rule application **322**. Other examples of application programs include calendars, search programs, email client applications, IM applications, SMS applications, Voice Over Internet Protocol (VOIP) applications, contact managers, task managers, transcoders, database programs, word processing programs, security applications, spreadsheet programs, games, search programs, and so forth.

(66) Extraction rule application **322** may be configured to enable creation of extraction rules and to display results of the extraction rules to a user. In at least one embodiment, extraction rule application **322** may interact with and/or employed through a web browser. In at least one of various embodiments, extraction rule application **322** may be configured to enable a user to select an emphasized (e.g., dimmed, highlighted, or the like) area within an event record and display each extraction rule, or field defined by an extraction rule, associated with the emphasized area. In some embodiments, embodiments, extraction rule application **322** may enable a user to input and/or edit one or more extraction rules. In other embodiments, extraction rule application **322** may display a plurality of event records to a user, values extracted from the event records using the extraction rule, statistics about the extracted values, or the like. In any event, extraction rule application **322** may employ processes, or parts of processes, similar to those described in conjunction with FIGS. **5-7**, to perform at least some of its actions.

(67) Illustrative Network Device

(68) FIG. **4** shows one embodiment of network device **400** that may be included in a system implementing the invention. Network device **400** may include many more or less components than those shown in FIG. **4**. However, the components shown are sufficient to disclose an illustrative embodiment for practicing the present invention. Network device **400** may represent, for example, one embodiment of at least one of network device **112**, **114**, or **120** of FIG. **1**.

(69) As shown in the figure, network device **400** may include a processor **402** in communication with a memory **404** via a bus **428**. Network device **400** may also include a power supply **430**, network interface **432**, audio interface **456**, display **450**, keyboard **452**, input/output interface **438**, processor-readable stationary storage device **434**, processor-readable removable storage device **436**, and pointing device interface **458**. Power supply **430** provides power to network device **400**.

(70) Network interface **432** may include circuitry for coupling network device **400** to one or more networks, and is constructed for use with one or more communication protocols and technologies

including, but not limited to, protocols and technologies that implement any portion of the Open Systems Interconnection model (OSI model), GSM, CDMA, time division multiple access (TDMA), UDP, TCP/IP, SMS, MMS, GPRS, W AP, UWB, WiMax, SIP/RTP, or any of a variety of other wired and wireless communication protocols. Network interface **432** is sometimes known as a transceiver, transceiving device, or network interface card (NIC). Network device **400** may optionally communicate with a base station (not shown), or directly with another computing device.

(71) Audio interface **456** is arranged to produce and receive audio signals such as the sound of a human voice. For example, audio interface **456** may be coupled to a speaker and microphone (not shown) to enable telecommunication with others and/or generate an audio acknowledgement for some action. A microphone in audio interface **456** can also be used for input to or control of network device **400**, for example, using voice recognition.

(72) Display **450** may be a liquid crystal display (LCD), gas plasma, electronic ink, light emitting diode (LED), Organic LED (OLED) or any other type of light reflective or light transmissive display that can be used with a computing device. Display **450** may be a handheld projector or pico projector capable of projecting an image on a wall or other object.

(73) Network device **400** also may also comprise input/output interface **438** for communicating with external devices not shown in FIG. **4**. Input/output interface **438** can utilize one or more wired or wireless communication technologies, such as USB™, Firewire™, WiFi, WiMax, Thunderbolt™, Infrared, Bluetooth™, Zig bee™, serial port, parallel port, and the like.

(74) Human interface components can be physically separate from network device **400**, allowing for remote input and/or output to network device **400**. For example, information routed as described here through human interface components such as display **450** or keyboard **452** can instead be routed through the network interface **432** to appropriate human interface components located elsewhere on the network. Human interface components can include any component that allows the computer to take input from, or send output to, a human user of a computer.

(75) Memory **404** may include RAM, ROM, and/or other types of memory. Memory **404** illustrates an example of computer-readable storage media (devices) for storage of information such as computer-readable instructions, data structures, program modules or other data. Memory **404** may store BIOS **408** for controlling low-level operation of network device **400**. The memory may also store operating system **406** for controlling the operation of network device **400**. It will be appreciated that this component may include a general-purpose operating system such as a version of UNIX, or LINUX™, or a specialized operating system such as Microsoft Corporation's Windows® operating system, or the Apple Corporation's iOS® operating system. The operating system may include, or interface with a Java virtual machine module that enables control of hardware components and/or operating system operations via Java application programs.

(76) Memory **404** may further include one or more data storage **410**, which can be utilized by network device **400** to store, among other things, applications **420** and/or other data. For example, data storage **410** may also be employed to store information that describes various capabilities of network device **400**. The information may then be provided to another device based on any of a variety of events, including being sent as part of a header during a communication, sent upon request, or the like. Data storage **410** may also be employed to store social networking information including address books, buddy lists, aliases, user profile information, or the like. Data stores **410** may further include program code, data, algorithms, and the like, for use by a processor, such as processor **402** to execute and perform actions. In one embodiment, at least some of data store **410** might also be stored on another component of network device **400**, including, but not limited to, non-transitory media inside processor-readable removable storage device **436**, processor-readable stationary storage device **434**, or any other computer-readable storage device within network device **400**, or even external to network device **400**.

(77) Data storage **410** may include, for example, event records **412** and extraction rules **416**. In

some embodiments, event records **412** may store data, including a plurality of event records. In at least one of various embodiments, event records **412** may be stored by event records server device **114** of FIG. **1**. Extraction rules **416** may include one or more extractions rules. These extraction rules may be automatically created based on a user selection of text, input by a user, and/or otherwise provided to the system. In at least one embodiment, extraction rules **416** may be stored and/or otherwise processed by extraction rule server device **112** of FIG. **1**.
(78) Applications **420** may include computer executable instructions which, when executed by network device **400**, transmit, receive, and/or otherwise process messages (e.g., SMS, MMS, Instant Message (IM), email, and/or other messages), audio, video, and enable telecommunication with another user of another client device. Other examples of application programs include calendars, search programs, email client applications, IM applications, SMS applications, Voice Over Internet Protocol (VOIP) applications, contact managers, task managers, transcoders, database programs, word processing programs, security applications, spreadsheet programs, games, search programs, and so forth. Applications **420** may include, for example, extraction rule application **422**.
(79) Extraction rule application **422** may be configured to enable creation of extraction rules and to display results of the extraction rules to a user. In at least one embodiment, extraction rule application **422** may interact with a client device for enabling a user to input and/or edit one or more extraction rules. In at least one of various embodiments, extraction rule application **322** may be configured to enable a user to select an emphasized (e.g., dimmed, highlighted, or the like) area within an event record and display each extraction rule, or field defined by an extraction rule, associated with the emphasized area. In other embodiments, extraction rule application **422** may enable a client device to display a plurality of event records to a user, values extracted from the event records using the extraction rule, statistics about the extracted values, or the like. In at least one embodiment, extraction rule application **422** may interact with event records **412** and/or extraction rules **416** to access and/or store event records and/or extraction rules, respectively. In some embodiments, extraction rule application **422** may be employed by extraction rule server device **112** of FIG. **1**. In any event, extraction rule application **422** may employ processes, or parts of processes, similar to those described in conjunction with FIGS. **5-7**, to perform at least some of its actions.
(80) General Operation
(81) The operation of certain aspects of the invention will now be described with respect to FIGS. **5-7**. FIG. **5** illustrates a logical flow diagram generally showing one embodiment of an overview process for enabling real time display of fields based on previously provided extraction rules;. In some embodiments, process **500** of FIG. **5** may be implemented by and/or executed on a single network device, such as network device **400** of FIG. **4**. In other embodiments, process **500** or portions of process **500** of FIG. **5** may be implemented by and/or executed on a plurality of network devices, such as network device **400** of FIG. **4**. In yet other embodiments, process **500** or portions of process **500** of FIG. **5** may be implemented by and/or executed on one or more blade servers, such as blade server **250** of FIG. **2**B. However, embodiments are not so limited and various combinations of network devices, blade servers, or the like may be utilized.
(82) Process **500** begins, after a start block, at block **502**, where a plurality of event records may be provided. In some embodiments, the event records may be provided by a plurality of different computing devices, such as client devices. In at least one embodiment, the plurality of event records may be a sample subset of a larger dataset of event records dataset. In some embodiments, the larger dataset of event records may be associated with one or more users and/or clients. As described above, the event records may be structured data or unstructured data. Additionally, the event records may include machine data.
(83) Process **500** proceeds next to block **504**, where data field extraction rules may be provided. In some embodiments, a plurality of extraction rules may be provided. The provided extraction rules

may define a field within the plurality of event records from which to extract data (e.g., a field value). Accordingly, in some embodiments, the extraction rule may define a field within the event records independent of a predetermined and/or predefined structure of the event records. Extraction rules may be provided independent of one another. In at least one of various embodiments, two or more extraction rules may define fields that may be distinct and/or separate fields. In other embodiments, two or more extraction rules may define fields that partially or completely overlap each other.

(84) In some embodiments, where fields overlap, an extraction rule may define a subfield of another field. In at least one embodiment, the other field may be defined by another extraction rule and/or may be a structured and/or predefined field. For example, Extraction Rule A may define a field as “Server_ID”, which may include a name of a server and an address of the server. Additionally, Extraction Rule B may define a field as “Server_name”, which may include the name of the server, but not the address of the server. In this example, Extraction Rule B may define a subfield of the field defined by Extraction Rule A; or Extraction Rule B may be referred to as a sub-rule to Extraction Rule A.

(85) In various embodiments, one or more extraction rules may be provided. Extraction rules may be automatically generated, manually entered by a user, previously provided/created, provided by another system, or the like, or any combination thereof. In at least one embodiment, automatic generation of an extraction rule may be based on a value selected from an event record. In some embodiments, a graphical user interface (GUI) may be employed to enable a user to select desired text of an event record. From the selected text, pattern recognition algorithms may be employed to automatically generate the extraction rule. In at least one embodiment, the extraction rule may be a regular expression.

(86) In another embodiment, the GUI may be employed to enable the user to manually input the extraction rule. In at least one embodiment, the user may enter a regular expression or other extraction rule into an editable input text box in the GUI to define a field within the event records from which to extract data.

(87) In yet other embodiments, the user may utilize the GUI to manually edit extraction rules (either previously automatically generated extraction rules or previous user-entered extraction rules) and receive a real time display of newly extracted values, statistics that correspond to the extracted values, changes to a display of the event records, or the like or any combination thereof. Real time display of field values based on manual editing of extraction rules is described in more detail below in conjunction with FIG. **6**.

(88) In some embodiments, the GUI may be employed to enable a user to provide a field name for the extraction rule (e.g., the field defined by the extraction rule). In other embodiments, the system may automatically determine a field name for the extraction rule. In at least one such embodiment, the system may employ the extraction rule to extract a value from one or more event records. The field name may be determined based on this value, such as, for example, a datatype of the extracted value (e.g., an integer), a format of the extracted value (e.g., a phone number, URL, time/date format), or the like. In various embodiments, the extraction rule may be automatically generated, manually input by a user, or the like, or any combination thereof.

(89) In any event, process **500** continues next at block **506**, where the GUI may be employed to display the event records based on the provided extraction rules in real time. In at least one embodiment, the plurality of event records may be displayed to the user in virtually any order, such as, most recent, latest, or the like.

(90) An embodiment of a process for displaying event records based on previously provided extraction rules is described in more detail below in conjunction with FIG. **6**. Briefly, however, in at least one embodiment, displaying an event record based on an extraction rule may include emphasizing the fields defined by the extraction rules (e.g., the extracted value) in the event record. Examples of such emphasizing may include, but are not limited to, dimming, highlighting,

underlining, balding, striking through, italicizing, displaying different font, displaying different font size, displaying different color, displaying different transparency, including parenthesis around the text, and the like. FIGS. **8**B and **8**C illustrate embodiments of real time display of event records where values associated with one or more fields defined by one or more extraction rules are emphasized.

(91) In some other embodiments, fields defined by different extraction rules may be emphasized in a same way or different ways. For example, in one embodiment, text of each defined field may be emphasized by displaying the text in a single font color. However, such emphasizing may make it difficult for a user to distinguish between fields or to determine if multiple fields overlap. In some other embodiments, each field may be emphasized differently. For example, in one embodiment, text of one defined field may be emphasized by displaying the text in one font, and text of a different defined field may be emphasized by displaying this text in a different font. However, embodiments are not so limited and other types of display emphasizing may be employed.

(92) In some embodiments, real time display of the event records may include displaying the event records based on the provided extraction rules as the extraction rules are being provided, entered, and/or edited by a user. Accordingly, the GUI may update a display of each event record and an indication of each extracted value in near real time as an extraction rule is edited/generated. It should be understood that real time or near real time display of data, as used herein, may include a delay created by some processing of the data, such as, but not limited to, a time to obtain an extraction rule, a time to determine text to emphasize based on the extraction rules, or the like.

(93) Process **500** proceeds next at block **508**, where a portion of at least one event record may be selected. The portion of the event record may include a subset, part, and/or area of a displayed event record. For example, in at least one of various embodiments, the portion may be a string of one or more characters, numbers, letters, symbols, white spaces, or the like. However, the selected portion is not limited to a subset of the displayed event record, but in another embodiment, the portion may include the entire displayed event record. In some other embodiments, the portion may span multiple event records.

(94) In some embodiments, the portion may include one or more fields defined by one or more extraction rules. In at least one such embodiment, the portion may be an emphasized area of the event record, such as fields that are emphasized in each event record (e.g., as described at block **506**). For example, text of an event record may be emphasized because that text is associated with at least one field defined by at least one extraction rule. In this example, the portion selected by the user may be the emphasized text. FIG. **8**C illustrates an embodiment of emphasized portions of an event record based on previously provided extraction rules.

(95) In at least one of various embodiments, a GUI may be employed to enable a user to select the portion of the event record. The user may select the portion of the event record by clicking on the portion of the event record, highlighting text of an event record, rolling over or mousing-over an area of the event record, or the like. For example, in at least one embodiment, a user may click on an emphasized portion of an event record to select it. In another embodiment, the user may roll a pointer over the emphasized portion of the event record to select it. In yet other embodiments, the user may utilize a text selection mechanism to highlight and select text of the event record to be the selected portion of the event record. These embodiments are non-limiting and non-exhaustive and other mechanisms may be employed to enable a user to select a portion of at least one event record. FIGS. **9**A-**9**B illustrate embodiments of a GUI being employed to enable a user to select a portion (e.g., emphasized fields) of an event record.

(96) Process **500** continues at block **510**, where extraction rules associated with the selected portion may be displayed, which is described in more detail below in conjunction with FIG. **7**. Briefly, however, in at least one of various embodiments, a window or pop-up box may open to display the associated extraction rules. In some embodiments, a name of the associated extraction rules may be displayed. In at least one such embodiment, this name may be a name of the field defined by the

extraction rule. In other embodiments, a value of each field defined by the extraction rule may be displayed. In at least one such embodiment, these values may be values extracted from the event record (from which the portion was selected to determine the associated extraction rules) using the associated extraction rules.

(97) In any event, process **500** proceeds to decision block **514**, where a determination may be made whether another portion of an event record is selected. In at least one embodiment, a user may select another portion of a same or different event record. Embodiments of block **508** may be employed to receive a selection of another portion of an event record. If another portion is selected, then process **500** may loop to block **510** to display extraction rules associated with the other selected portion; otherwise, process **500** may return to a calling process to perform other actions.

(98) FIG. **6** illustrates a logical flow diagram generally showing one embodiment of a process for displaying event records that emphasizes fields based on previously provided extraction rules. In some embodiments, process **600** of FIG. **6** may be implemented by and/or executed on a single network device, such as network device **400** of FIG. **4**. In other embodiments, process **600** or portions of process **600** of FIG. **6** may be implemented by and/or executed on a plurality of network devices, such as network device **400** of FIG. **4**. In yet other embodiments, process **600** or portions of process **600** of FIG. **6** may be implemented by and/or executed on one or more blade servers, such as blade server **250** of FIG. **2**B. However, embodiments are not so limited and various combinations of network devices, blade servers, or the like may be utilized.

(99) Some markup languages, such as HTML or XML, do not allow overlapping tag pairs. This type of limitation can make it difficult to display individual fields that overlap one another, where each field may be defined by a tag pair that may overlap another tag pair. Process **600** describes embodiments for displaying overlapping and/or sub-containing sections of text (e.g., overlapping fields and/or sub-fields) within an overlapping tag-pair-limited mark-up language, such as, but not limited to HTML or XML. Process **600** further describes embodiments that enable the display of overlapping fields while preserving individual information segments (e.g., field values) contained within each field or tag pair.

(100) Process **600** begins, after a start block, at block **602**, where an event record may be selected. In at least one embodiment, event records may be randomly selected from a plurality of event records (e.g., the plurality of event records provided at block **502** of FIG. **5**). In another embodiment, event records may be selected in a predetermined order, such as chronologically (e.g., based on a timestamp), reverse chronologically, alphabetically, or the like. In yet other embodiments, a field, such as a field defined by an extraction rule, may be utilized to determine an order of selecting event records. For example, a field may define a server identifier and event records may be selected based on the server identifier. However, other mechanisms and/or algorithms may be employed for determining which event record to select.

(101) Process **600** proceeds at block **604**, where an extraction rule may be selected. In at least one embodiment, the extraction rule may be selected from a plurality of extraction rules that were previously provided (e.g., created, stored, or the like). The plurality of extraction rules may have been automatically generated, manually created, or the like, such as is described at block **504** of FIG. **5**.

(102) Process **600** continues at block **606**, where a field defined by the selected extraction rule may be determined. In at least one embodiment, this determination may include using the selected extraction rule to determine and/or identify text and/or a value of the selected event record that corresponds to the field defined by the selected extraction rule. In some embodiments, this text and/or value (or a location and size of this text/value within the selected event record) may be at least temporarily maintained/stored and used to display the selected event record at block **610**.

(103) In any event, process **600** proceeds to decision block **608**, where a determination may be made whether another extraction rule may be selected. In some embodiments, another extraction rule may be selected from a plurality of extraction rules until each of the plurality of extraction

rules is selected. If another extraction rule may be selected, then process **600** may loop to block **604** to select another extraction rule; otherwise, process **600** may flow to block **610**.

(104) At block **610**, the selected event record may be displayed with an emphasis of each determined field (e.g., as determined at block **606**). As described above, in at least one embodiment, a display of text of each determined field may be emphasized within the selected event record. In some embodiments, each determined field may be emphasized in the same way, such as, for example, all may be emphasized with a light blue highlight. In other embodiments, each determined field may be emphasized in a different way, such as, for example, each determined field may be enclosed in different colored parentheses. However, embodiments are not so limited, and other mechanisms for emphasizing the determined fields in the selected event record may be employed.

(105) In some embodiments, two or more determined fields may overlap. In at least one such embodiment, the corresponding text/values may be combined and emphasized together as a super set field, such that each overlapping field may not be individually distinguished from one another. Accordingly, in some embodiments, the combined text may be employed to emphasize a plurality of fields in a super set field that is defined by a plurality of different extraction rules.

(106) In at least one embodiment, a start and end character location of the determined fields within the selected event record may be utilized to determine if fields overlap. For example, assume in the selected event record, Field_A has a start character location of 5 and an end character location of 10 and Field_B has a start character location of 7 and an end character location of 15. In this example, a combined text from character location 5 to 15 may be emphasized.

(107) In some other embodiments, the start and end character location of multiple determined fields may be compared to determine a super set or most inclusive field. For example, assume the above example is expanded to include Field_C that has a start character location of 5 and an end character location of 22. In this expanded example, the combined text that may be emphasized may be from character location 5 to 22. Additionally, in this expanded example, Field_A and Field_B may be sub-fields of Field_C (and may or may not be sub-fields of each other). One example of a code fragment for determining a super set field that contains continuous fields is illustrated in FIGS. **1** OA-**1** OC. This code fragment also stores sub-field information for display on rollover (such as may be utilized at blocks **708**, **710**, and/or **712** of FIG. **7**).

(108) In any event, process **600** continues next at decision block **612**, where a determination may be made whether another event record may be selected. In some embodiments, another event record may be selected from a plurality of event records until each of the plurality of event records is selected and displayed. If another event record may be selected, then process **600** may loop to block **602** to select another event record; otherwise, process **600** may return to a calling process to perform other actions.

(109) FIG. **7** illustrates a logical flow diagram generally showing one embodiment of a process for displaying previously provided extraction rules associated with a selected portion of an event record. In some embodiments, process **700** of FIG. **7** may be implemented by and/or executed on a single network device, such as network device **400** of FIG. **4**. In other embodiments, process **700** or portions of process **700** of FIG. **7** may be implemented by and/or executed on a plurality of network devices, such as network device **400** of FIG. **4**. In yet other embodiments, process **700** or portions of process **700** of FIG. **7** may be implemented by and/or executed on one or more blade servers, such as blade server **250** of FIG. **2**B. However, embodiments are not so limited and various combinations of network devices, blade servers, or the like may be utilized.

(110) Process **700** begins, after a start block, at block **702**, where a portion of an event record may be selected. In at least one of various embodiments, block **702** may employ embodiments of block **508** to select a portion of an event record.

(111) Process **700** proceeds to decision block **704**, where a determination may be made whether there is one or more extraction rules associated with the selected portion that was not previously

selected at block **706**. In some embodiments, process **700** may proceed through blocks **706**, **708**, **710**, and **712** once for each extraction rule associated with the selected portion. If one or more extraction rules are associated with the selected portion, then process **700** may flow to block **706**; otherwise, process **700** may return to a calling process to perform other actions.

(112) At block **706**, an extraction rule associated with selected portion may be selected. In at least one embodiment, the selection of an extraction rule may be random, in a predetermined order, or the like.

(113) Process **708** proceeds next to block **708**, where an identifier of the selected extraction rule may be displayed. In some embodiments, this identifier may include a name of the field defined by the selected extraction rule. In other embodiments, this identifier may be an extraction rule name. In yet other embodiments, the selected extraction rule itself may be displayed.

(114) Process **700** continues at block **710**, where the selected extraction rule may be used to extract a value from the event record from which the selected portion was selected. In at least one of various embodiments, the selected extraction rule may be applied to the event records to determine data to extract from the event record. The extracted data from the event record may be the particular value for the event record for the field defined by the selected extraction rule. For example, if the selected extraction rule defines a field as the characters between a first set of single brackets, then the value for the event record “December 17 10:35:38 ronnie nslcd[23629]: [40f750] passwd entry uid” may be “23629”.

(115) In any event, process **700** proceeds at block **712**, where the extracted value may be displayed. In at least one embodiment, the extracted value may be displayed next to or in conjunction with the identifier of the selected extraction rule. An example of a GUI displaying an identifier of the selected extraction rule and a corresponding extracted value is illustrated in FIGS. **9**A-**9**B.

(116) After block **712**, process **700** may loop to decision block **704** to determine if there may be another extraction rule associated with the selected portion that was not previously selected at block **706**.

(117) It will be understood that each block of the flowchart illustration, and combinations of blocks in the flowchart illustration, can be implemented by computer program instructions. These program instructions may be provided to a processor to produce a machine, such that the instructions, which execute on the processor, create means for implementing the actions specified in the flowchart block or blocks. The computer program instructions may be executed by a processor to cause a series of operational steps to be performed by the processor to produce a computer-implemented process such that the instructions, which execute on the processor to provide steps for implementing the actions specified in the flowchart block or blocks. The computer program instructions may also cause at least some of the operational steps shown in the blocks of the flowchart to be performed in parallel. Moreover, some of the steps may also be performed across more than one processor, such as might arise in a multi-processor computer system. In addition, one or more blocks or combinations of blocks in the flowchart illustration may also be performed concurrently with other blocks or combinations of blocks, or even in a different sequence than illustrated.

(118) Accordingly, blocks of the flowchart illustration support combinations of means for performing the specified actions, combinations of steps for performing the specified actions and program instruction means for performing the specified actions. It will also be understood that each block of the flowchart illustration, and combinations of blocks in the flowchart illustration, can be implemented by special purpose hardware-based systems, which perform the specified actions or steps, or combinations of special purpose hardware and computer instructions.

(119) Use Case Illustration

(120) FIGS. **8**A-**8**C illustrate non-exhaustive examples of use case of embodiments of a graphical user interface that may be employed to enable a user to create extraction rules and to display indications of previously extracted data fields.

(121) FIG. **8**A illustrates a non-exhaustive example of a use case of an embodiment of graphical user interface that may be employed to enable a user to create extraction rules. Graphical user interface (GUI) **800**A may include multiple viewing windows and/or sections that each display information to a user. For example, GUI **800**A may include records **808**, input **802**, input **806**, extraction rule preview **804**, records **808**, and extracted values **810**.
(122) Records **808** may display each event record that is determined based on inputs **802** and **806**. Input **802** may enable a user to input a data source (e.g., a specific database) and/or a data type (e.g., system log data). As illustrated, input **802** may include one or more pull down menus of available options of the data sources and/or data types available. However, other menus, lists, windows, or interfaces may also be employed. Input **806** may enable the user to define a specific filter to apply the event records (e.g., the user may filter the event records to display those event records that were recorded on a particular day). In other embodiments, input **806** may enable a user to select how the event records are selected for display. In at least one embodiment, event records **808** may include a subset and/or sampling of a lager data set. For example, input **806** may be used to select that event records **808** includes a predetermined number (e.g., 1 00) of the latest event records. However, other result types may be used, such as oldest, most popular, least popular, or the like, or any combination thereof.
(123) Extraction rule preview **804** may display instructions to a user for creating an extraction rule. For example, the user may highlight and/or select text in an event record in records **808** to have an extraction rule automatically created. In another example, the user may manually enter an extraction rule (e.g., by clicking on the “Create extraction rule” button, an editable text box may open or become visible where the user can manually input an extraction rule). Extraction rule preview **804** may display the extraction rule after it is created, such as is shown in FIG. **8**B. Additionally, the user may be enabled to save the extraction rule for additional processing of event records and extracted values.
(124) Extracted values **810** may show unique values that are extracted from event records **808** based on an extraction rule provided by extraction rule preview **804**. As illustrated, extracted values **810** may be empty because no extraction rule has been provided.
(125) FIG. **8**B illustrates a non-exhaustive example of a use case of an embodiment of a graphical user interface where an extraction rule has been provided. GUI **800**B may be an embodiment of GUI **800**A from FIG. **8**A.
(126) Extraction rule preview **804** may display the provided extraction rule. In at least one embodiment, GUI **800**B may include editable text box **814** to enable the user to provide a field name of the field defined by the extraction rule. As described above, the extraction rule may have been automatically generated based on user selected text from an event record in the event records **808**. In other embodiments, a user may have manually entered the extraction rule. As illustrated, the extraction rule may be displayed in editable text box **812**. Editable text box **812** may enable a user to manually edit the extraction rule. As the user is manually editing the extraction rule, records **808** may be automatically and dynamically updated in real time to show new values extracted from each event record in records **808**. For example, the extracted values from each event record may be emphasized, as shown by emphasis **824**. Additionally, extracted values **810** may be automatically and dynamically updated in real time as the user edits the extraction rule.
(127) In other embodiments, the extraction rule may be manipulated by indicating an incorrect extracted value (e.g., a counter-example). In at least one embodiment, a counter-example may be a value extracted from an event record based on an extraction rule that does not match a desired field of the user. For example, assume an extraction rule is created to define a field for a server name. However, assume the extraction rule extracts other data from at least one of the event records. The user may indicate this other data as a counter-example, and the system may automatically regenerate the extraction rule taking this counter-example into account. In at least one of various embodiments, a user may indicate a counter-example by clicking on a counter-example button,

such as button **822**. By clicking button **822**, the system may automatically re-generate the extraction rule based on the counter example and the other extracted values.

(128) Extracted values **810** may include one or more unique values extracted from records **808** based on the extraction rule. In at least one embodiment, statistics that correspond to each unique extracted value may be displayed. For example, data **816** shows a percentage of the number of times each particular unique value is extracted from records **808**. As illustrated, each of these percentages may also be illustrated as a percentage bar (e.g., percentage bar **818**) for each unique extracted value.

(129) FIG. **8**C illustrates a non-exhaustive example of a use case of an embodiment of graphical user interface that may be employed to display event records with an emphasis of fields defined by previously provided extraction rules. GUI **800**C may be an embodiment of GUI **800**A of FIG. **8**A.

(130) GUI **800**C may include input **826**. Input **826** may be a check box or other mechanism that may be selected by a user. In at least one embodiment, a selection of input **826** may display records **808** with emphasized fields defined by previous extraction rules. As illustrated, each event record in records **808** may include one or more emphasized sections of text, such as sections **828** and **830**. In some embodiments, an emphasized section, such as section **830**, may include a plurality of at least partially overlapping fields. As shown, these overlapping fields may not be distinguished from one another. However, in other embodiments (not shown), these overlapping fields may be distinguished from one another using different types of emphasis.

(131) FIGS. **9**A-**9**B illustrate non-exhaustive examples of a use case of embodiments of a graphical user interface, such as depicted in FIGS. **8**A-**8**C, to display extraction rules and/or fields associated with a selected portion of an event record.

(132) GUI **900**A may be an embodiment of GUI **800**C. As illustrated, a user may move a cursor or other pointer over section **904** to select section **904**. By selecting section **904**, GUI **900**A may display extraction rules associated with that portion of event record **920**. In at least one embodiment, section **904** may be an embodiment of section **828** of FIG. **8**C. By employing embodiments described above, a box **906** may pop-up and/or open to display an extraction rule that is associated with section **904** of event record **920**. In this example, box **906** may include a fieldname of a field defined by the associated extraction rule ("Server ID") and a value extracted from event record **920** using the associated extraction rule ("23629"). In this illustration section **904** may be associated with a single extraction rule.

(133) GUI **900**B may be an embodiment of GUI **900**A. As illustrated, a user may move a cursor or other pointer over section **910** to select section **910**. By selecting section **910**, GUI **900**B may display extraction rules associated with that portion of event record **920**. In at least one embodiment, section **910** may be an embodiment of section **830** of FIG. **8**C. By employing embodiments described above, box **912** may pop-up and/or open to display extraction rules that are associated with section **910** of event record **920**. In some embodiments, box **912** may be an embodiment of box **906** of FIG. **9**A. In this example, box **912** may include a fieldname of a field defined by each associated extraction rule and corresponding value extracted from event record **920** using the associated extraction rules. In this example illustration, section **910** may have three different extraction rules associated with it, and an identifier of each extraction rule may be displayed in box **912** (e.g., "Error", "Error type", and "User ID"). Additionally, each associated extraction rule may be used to extract a corresponding value from event record **920**, which may also be displayed in box **912**.

(134) Moreover, some fields may be sub-fields of other fields. In this example, fieldnames "Error type" and "User ID" may be sub-fields of fieldname "Error" because fieldname "Error" overlaps both fieldname "Error type" and "User ID".

(135) FIGS. **10**A-**10**C illustrate a non-exhaustive example of a code fragment for determining a part of an event record to emphasize based on a plurality of fields.

(136) The above specification, examples, and data provide a complete description of the

composition, manufacture, and use of the invention. Since many embodiments of the invention can be made without departing from the spirit and scope of the invention, the invention resides in the claims hereinafter appended.

Claims

1. A method, comprising: causing display, via a graphical user interface, of an extraction rule and a set of event records; receiving, via the graphical user interface, user input editing the extraction rule displayed via the graphical user interface, the extraction rule defining at least one field within the set of event records from which to extract a value; and causing display, via the graphical user interface, of an updated set of event records that includes an indication of at least one new value extracted based on the edited extraction rule.
2. The method of claim 1, wherein the event record comprises at least a portion of one or more lines of data within machine data.
3. The method of claim 1, wherein the extraction rule comprises a regular expression.
4. The method of claim 1, wherein the updated set of event records is updated in real time as the user input is received.
5. The method of claim 1, wherein the extraction rule displayed via the graphical user interface is generated based on a text value selected, via the graphical user interface, from an event record, wherein the extraction rule is generated using a pattern recognition algorithm.
6. The method of claim 1 further comprising automatically generating the extraction rule based on a value selected from an event record.
7. The method of claim 1, wherein the extraction rule is input, via the graphical user interface, into an editable input text box.
8. The method of claim 1, wherein the set of event records are displayed, via the graphical user interface, in real time based on the extraction rule.
9. The method of claim 1, wherein the at least one new value extracted is visually emphasized in the updated set of event records.
10. The method of claim 1 further comprising causing display of one or more unique values extracted from the updated set of event records based on the edited extraction rule.
11. The method of claim 1, further comprising: causing display of one or more unique values extracted from the updated set of event records based on the edited extraction rule; and causing display of one or more statistics that correspond to the one or more unique values.
12. The method of claim 1, further comprising receiving user input of a field name of the at least one field defined by the extraction rule.
13. The method of claim 1, wherein the extraction rule is selected from a plurality of extraction rules.
14. The method of claim 1, wherein the extraction rule is displayed along with a name of the extraction rule.
15. A computerized system comprising: one or more processors; and computer storage memory having computer-executable instructions stored thereon which, when executed by the processor, implement a method comprising: causing display, via a graphical user interface, of an extraction rule and a set of event records; receiving, via the graphical user interface, user input editing the extraction rule displayed via the graphical user interface, the extraction rule defining at least one field within the set of event records from which to extract a value; and causing display, via the graphical user interface, of an updated set of event records that includes an indication of at least one new value extracted based on the edited extraction rule.
16. The computerized system of claim 15, wherein the updated set of event records is updated in real time as the user input is received.
17. The computerized system of claim 15, wherein the extraction rule displayed via the graphical

user interface is generated based on a text value selected, via the graphical user interface, from an event record, wherein the extraction rule is generated using a pattern recognition algorithm.

18. A non-transitory computer-readable medium storing one or more sequences of instructions, wherein execution of the one or more sequences of instructions by one or more processors causes the one or more processors to perform a method comprising: causing display, via a graphical user interface, of an extraction rule and a set of event records; receiving, via the graphical user interface, user input editing the extraction rule displayed via the graphical user interface, the extraction rule defining at least one field within the set of event records from which to extract a value; and causing display, via the graphical user interface, of an updated set of event records that includes an indication of at least one new value extracted based on the edited extraction rule.

19. The non-transitory computer-readable medium of claim 18, wherein the set of event records are displayed, via the graphical user interface, in real time based on the extraction rule.

20. The method of claim 1, wherein the extraction rule is displayed based on a user selected portion of an event record in the displayed set of event records, and wherein the extraction rule is associated with the user selected portion of the event record.